

Day 6 lab: pruning, distillation, and model selection

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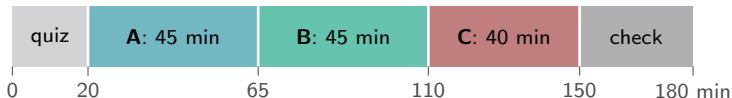
- ▶ **Goal:** your group has a trade-off curve for one task, and a model selection for the two boards of the course.
- ▶ **Boards:** none. The lab runs on the laptop.
- ▶ **Deliverable:** the plot `tradeoff.png` with at least six models, and the file `report.md` with your selection and the Decision Log.

From the lecture to the lab

The lecture gave three ways to a smaller model and a method for the selection. In the lab, you prune a CNN, you distill a teacher into a student, you quantize the candidates, and you select one model for each board from your own plot.

Key point: A method is good only for a given budget. The plot shows all candidates. The budget selects one of them.

Plan of the three hours



Part	Work	Time
Quiz	The quiz of Week 1	20 min
A	Pruning: prune a CNN at several sparsity levels and fine-tune it	45 min
B	Distillation: distill a teacher model into a smaller student model	45 min
C	Selection: quantize the candidates, plot, and select one model for each board	40 min
Check by the instructor		30 min

Write each result in `report.md` when you get it.

You need

- ▶ A laptop. No board.
- ▶ The Python environment of Days 1 and 3, with tensorflow-cpu and ai-edge-litert.
- ▶ No new package.

Lab files Labs/day06/. Run the commands from this folder.

Notebook `../../.venv/bin/jupyter lab compression.ipynb`

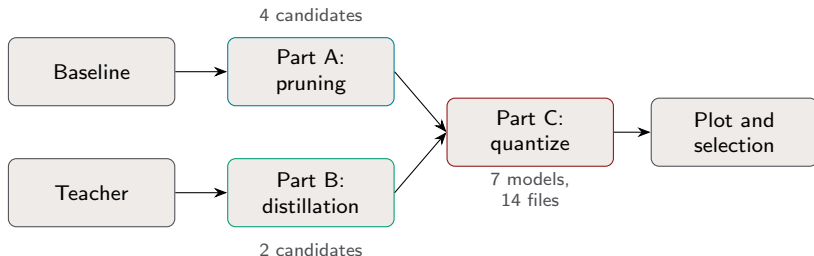
Dataset Fashion-MNIST. The first cell downloads it: about 30 MB.

- ▶ The two models are trained. A training from the start needs too much time for a lab.

The lab folder gives

- ▶ `compression.ipynb`: the notebook with seven tasks.
- ▶ `models/baseline.keras`: the CNN of the lecture.
- ▶ `models/teacher.keras`: a wider CNN.

From two models to one plot



- ▶ Each part adds models to one list of candidates. The baseline is the first candidate.
- ▶ Part C converts each candidate to a `float32` file and to an `int8` file, and it measures four numbers for each file: the size, the accuracy, the latency, and the peak of the activations.
- ▶ An experiment starts only when the check of its task prints complete.

Key point: The latency of this lab is the latency of your laptop. Day 9 measures the same models on the boards.

Task	You write	The notebook checks
1	<code>magnitude_mask</code> : the threshold and the mask for a sparsity	A matrix of 10 weights with a sparsity of 0.6: the 4 largest magnitudes stay
2	Your prediction for a sparsity of 80 percent	Not checked. You compare it with the result.
3	<code>filter_scores</code> : the sum of the absolute weights of each filter	A kernel with four known filters

$$\hat{W} = M \odot W \quad M_{ij} = 1 \text{ if } |W_{ij}| > \delta$$

- ▶ The threshold δ is the value that the given part of the magnitudes is below. Use `np.quantile`.
- ▶ A kernel has the shape (height, width, input channels, filters). The last axis is the filter.

Section 2 prunes single weights. Section 3 removes complete filters. Each model gets 2 epochs of fine-tuning.

sparsity	weights not 0	before	after	file	gzip	latency
none	23824		89.1	99320	...	...
0.50	...	...	...	...	...	...
filters	parameters	before	after	file	latency	peak
16-32-64	23946		89.1	99320	...	62720
12-24-48	...	...	...	...	...	...

- ▶ before and after: the accuracy before and after the fine-tuning.
- ▶ **You record:** the two tables, and the answers to the three questions of section 4.

Prediction (task 2). For a sparsity of 80 percent: which accuracy before and after the fine-tuning? Is the LiteRT file smaller, equal, or larger? Is the latency smaller, equal, or larger?

Source: The rows of the baseline are the values of the model file of the lab. The latency depends on your laptop.

Task	You write	The notebook checks
4	<code>soft_targets</code> : the softmax at a temperature T , in NumPy	The logits 4, 2, 0, and -2 of the lecture exercise, at $T = 1$ and $T = 4$
5	The three lines of the soft part of <code>distillation_loss</code>	The soft part is 0 for a student that is equal to the teacher
6	Your prediction for the four runs	Not checked. You compare it with the result.

$$L = \alpha \cdot L_{\text{hard}} + (1 - \alpha) \cdot T^2 \cdot L_{\text{soft}}$$
$$L_{\text{soft}} = \sum_i p_{\text{teacher},i}^T \cdot \left(\log p_{\text{teacher},i}^T - \log p_{\text{student},i}^T \right)$$

- ▶ Use `ops.softmax`, `ops.log_softmax`, `ops.log`, and `ops.sum` with `axis=-1`.
- ▶ The teacher has 93 962 parameters. The student has 6218.

Part B: four runs of one student

Run	Training data	Target
1	1000 labelled images: 100 for each class	The labels
2	The same 1000 images	The labels and the soft targets ($T = 2, \alpha = 0.5$)
3	All 60 000 images, with no label	The soft targets only ($T = 2, \alpha = 0$)
4	All 60 000 labelled images	The labels

- ▶ Runs 1 and 2: a team with few labels. Run 3: a team with a teacher and much raw data. Run 4: the reference. The four runs need about three minutes.
- ▶ **You record:** the four accuracy values, the two gains against run 1, and the answers to the three questions.

Prediction (task 6). How many points do runs 2 and 3 gain against run 1? Which run is better: run 3 or run 4?

Section 8 converts each candidate with the method of Day 4 and measures each file:

model	format	bytes	accuracy	latency	peak
baseline 16-32-64	float32	99320	89.1	...	62720
baseline 16-32-64	int8	...	...	...	...
pruned 12-24-48	float32	...	...	...	...

- ▶ **peak:** the largest sum of the input and the output of one operator, in bytes.
- ▶ Section 9 writes the plot `tradeoff.png`: the accuracy against the file size, and the accuracy against the latency.
- ▶ Find the files on the upper left border of each plot: no other file is smaller and more accurate.
- ▶ **You record:** the table, and the files on the border of each plot.

Look at your table. Is `int8` faster or slower than `float32` on your laptop? Compare with the group next to you.

Part C: select one model for each board

Product	Budget	Rule
XIAO ESP32S3	File of 16 KB or less. Peak of 8 KB or less. Format <code>int8</code> .	The highest accuracy that meets the budget
Raspberry Pi 5	Accuracy of 88.5 percent or more. Format <code>float32</code> or <code>int8</code> .	The smallest latency that meets the budget

- ▶ The budgets are example budgets: the model shares each board with other functions.
- ▶ **Task 7.** Write the model and the format for each product in the cell of the task. Name the constraint that decides: the budget that removes the next better model.
- ▶ The check prints, for each product, if your file meets the budget, and if a better file exists.

Key point: The two products get different models from the same plot. The constraint decides, not the model.

The instructor checks this at the end of the lab:

- ▶ ✓ The notebook prints `complete` for tasks 1, 3, 4, 5, and 7.
- ▶ ✓ The report has the two predictions: task 2 and task 6. You wrote them before the experiments.
- ▶ ✓ The plot `tradeoff.png` has at least six models.
- ▶ ✓ The report names one model for the XIAO ESP32S3 and one model for the Raspberry Pi 5, with the numbers of each model.
- ▶ ✓ The selection states the constraint that decides for each product.
- ▶ ✓ The Decision Log gives numbers and names one trade-off.

Hand in: the file `report.md` and the plot `tradeoff.png`.

Write about 100 words. State one design decision, give your measured numbers, and name the trade-off.

Which model do you select for the XIAO ESP32S3, and which model for the Raspberry Pi 5? Use the budgets of section 10 of the notebook.

A good Decision Log:

- ▶ gives the file size, the peak, the accuracy, and the latency of each selected model,
- ▶ names the constraint that decides for each product, and the model that this constraint removes,
- ▶ says what the product loses with the selection. Example: some points of accuracy against half of the RAM.

If a part does not work, and common problems

Problem	Fix or fallback
The download of the dataset fails	The instructor gives the folder <code>fashion-mnist</code> . Copy it into the folder <code>.keras/datasets/</code> of your home folder.
A task is not complete in time	Copy the function from the notebook of <code>solutions/</code> . Write this in the report.
An experiment prints that it does not run	Complete the task before it, run its check cell, then run the experiment.
Task 1: not complete after your change	The mask has the type <code>float32</code> . A weight stays only if its magnitude is above the threshold.
Task 5: not complete	Divide the two logit arrays by <code>T</code> before the softmax.
Section 8 has fewer than six models	Complete tasks 1, 3, and 5, and run the three experiments again.
The latency changes from run to run	Close other programs. Some microseconds are noise.
Task 7: not complete	Read the two lines that the check prints for each product.

The README of the lab has the complete list.

This lecture adapts material from these sources:

- ▶ *Machine Learning Systems* by Vijay Janapa Reddi and contributors, Harvard University. mlsysbook.ai, CC BY-NC-SA 4.0.
- ▶ *Edge AI Engineering: Raspberry Pi* by Marcelo Rovai. github.com/Mjrovai/EdgeML_Made_Ease_ebook and github.com/Mjrovai/EdgeML-with-Raspberry-Pi, GPL-3.0.

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